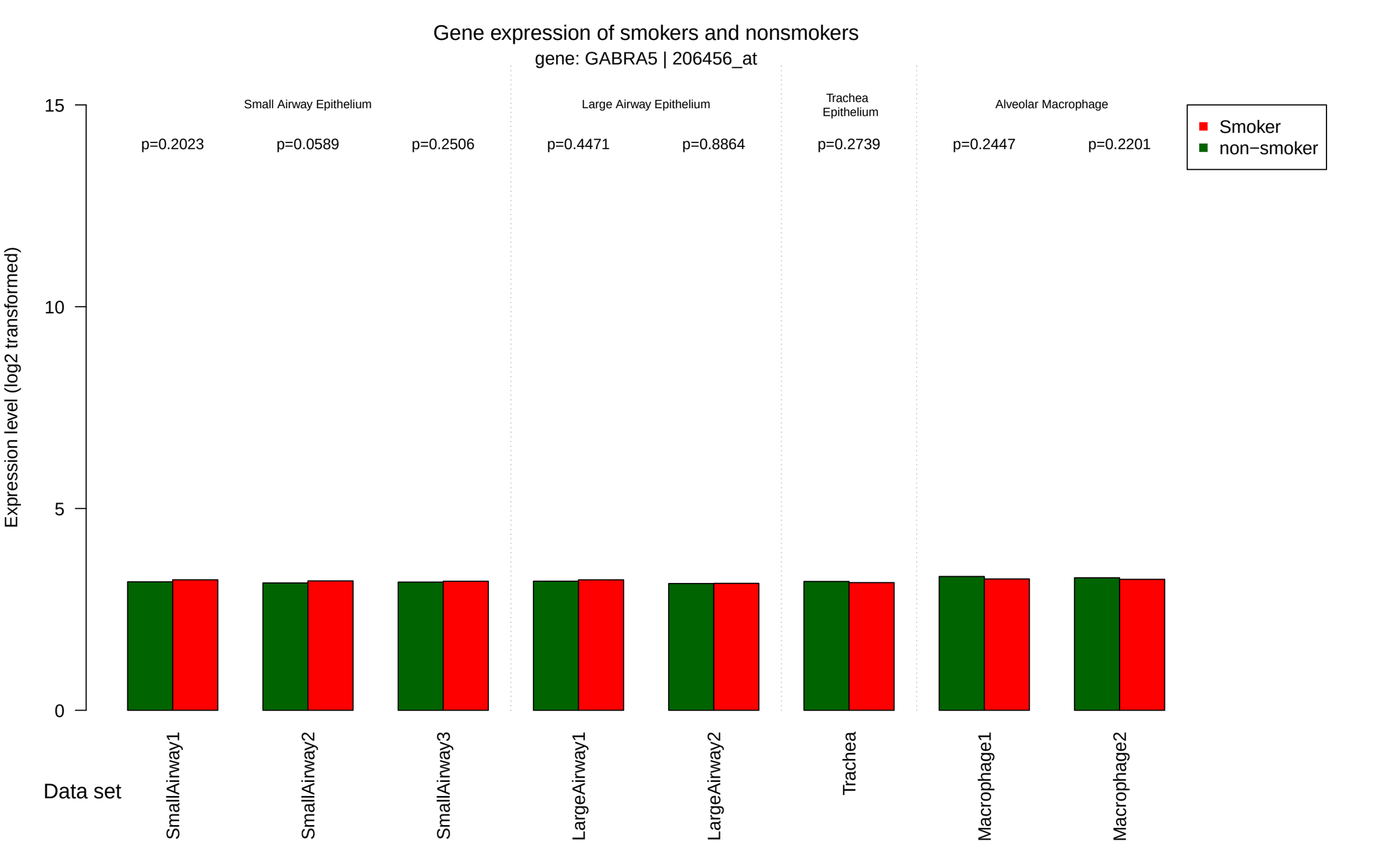
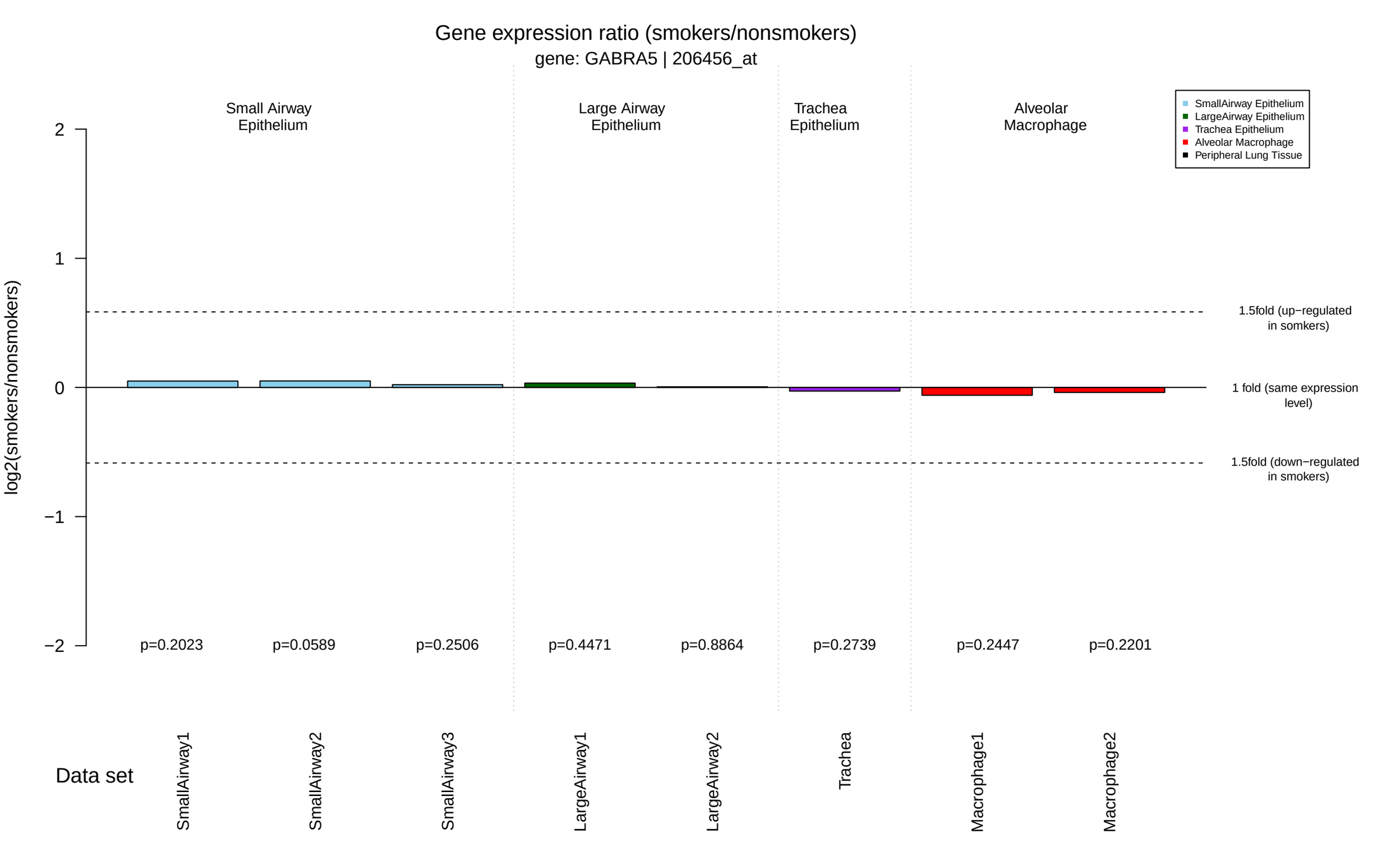
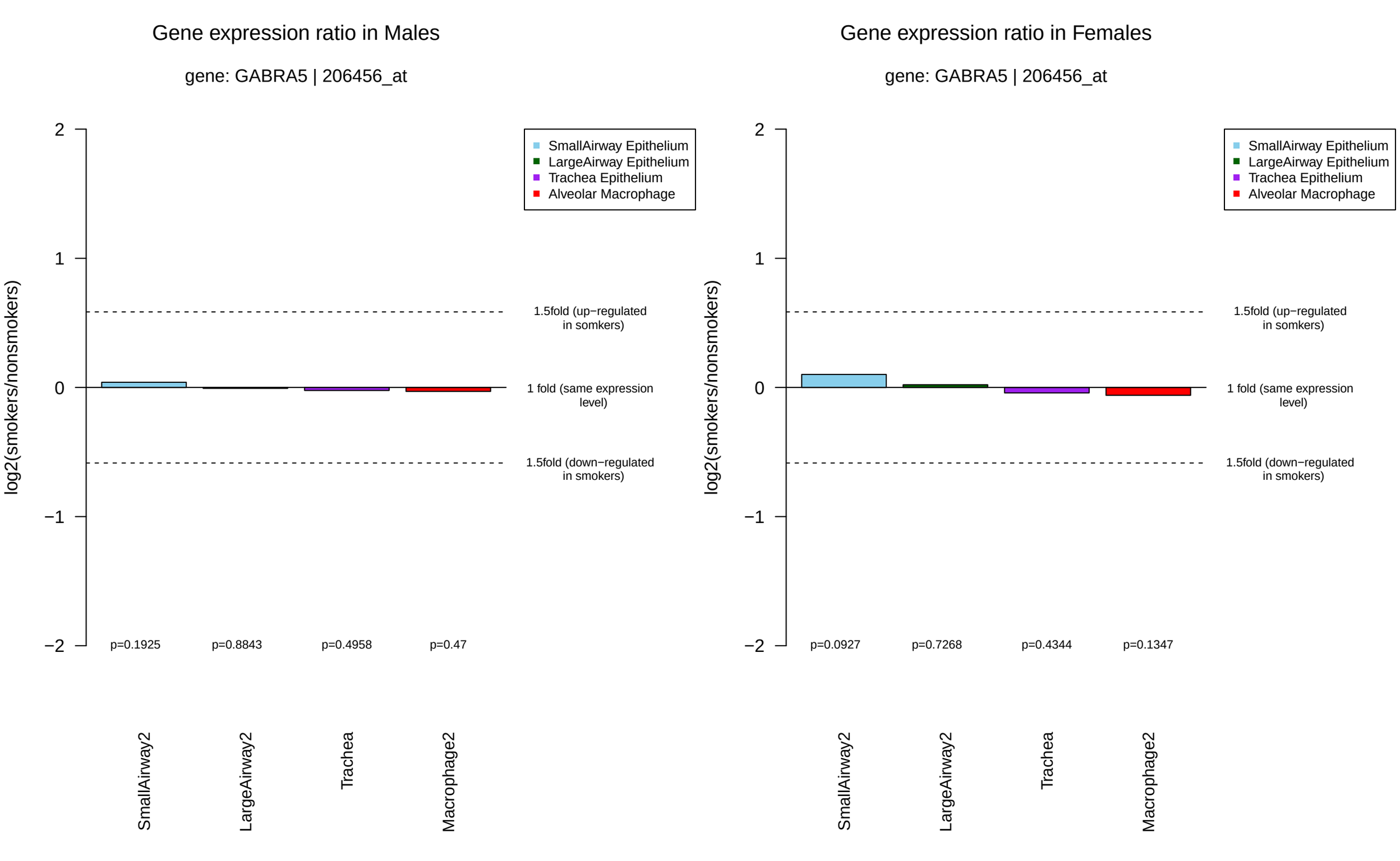
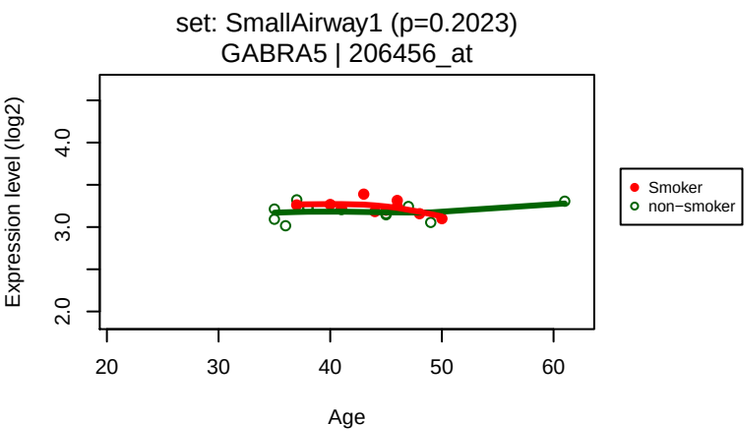
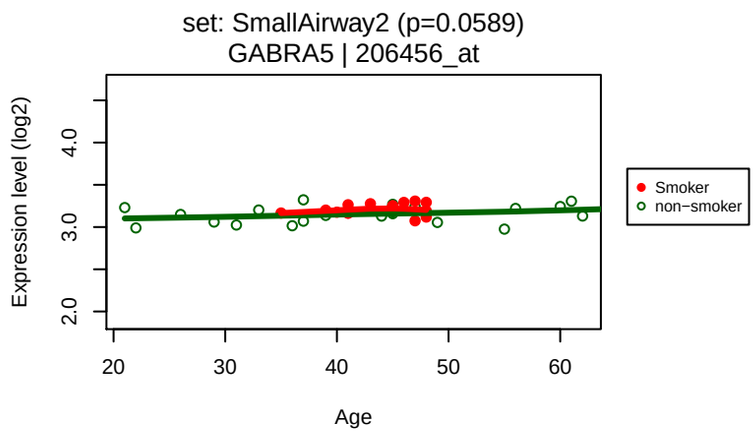
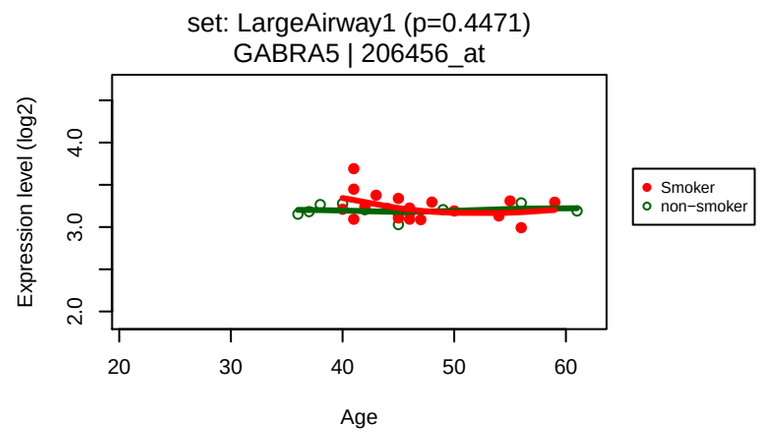

Gene expression of smokers and nonsmokers
gene: GABRA5 | 206456_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2023
p=0.0589
p=0.2506
p=0.4471
p=0.8864
p=0.2739
p=0.2447
p=0.2201
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

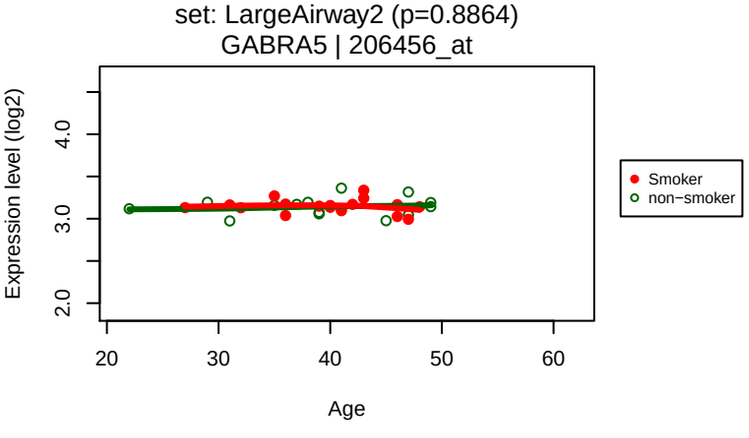

Gene expression ratio (smokers/nonsmokers)
gene: GABRA5 | 206456_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2023
p=0.0589
p=0.2506
p=0.4471
p=0.8864
p=0.2739
p=0.2447
p=0.2201
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

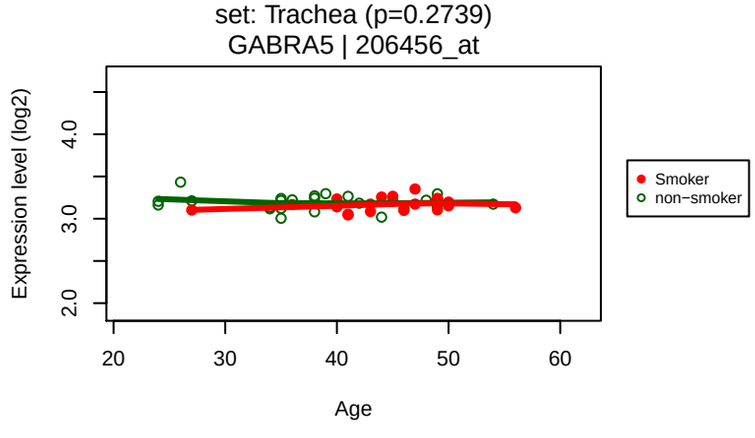

Gene expression ratio in Males
gene: GABRA5 | 206456_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1925
p=0.8843
p=0.4958
p=0.47
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRA5 | 206456_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0927
p=0.7268
p=0.4344
p=0.1347
SmallAirway2
LargeAirway2
Trachea
Macrophage2

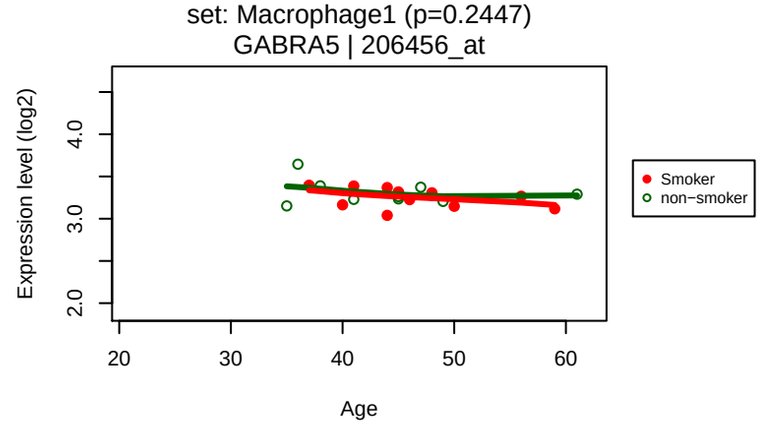

set: SmallAirway1 (p=0.2023)
GABRA5 | 206456_at
Expression level (log2)
Age
Smoker
non−smoker

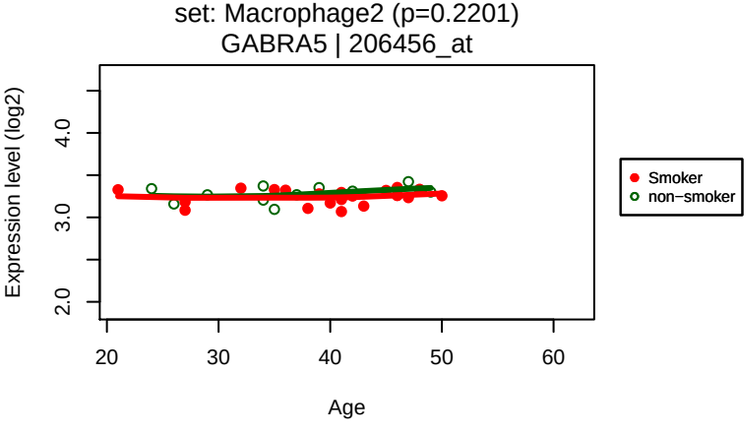

set: SmallAirway2 (p=0.0589)
GABRA5 | 206456_at
Expression level (log2)
Age
Smoker
non−smoker

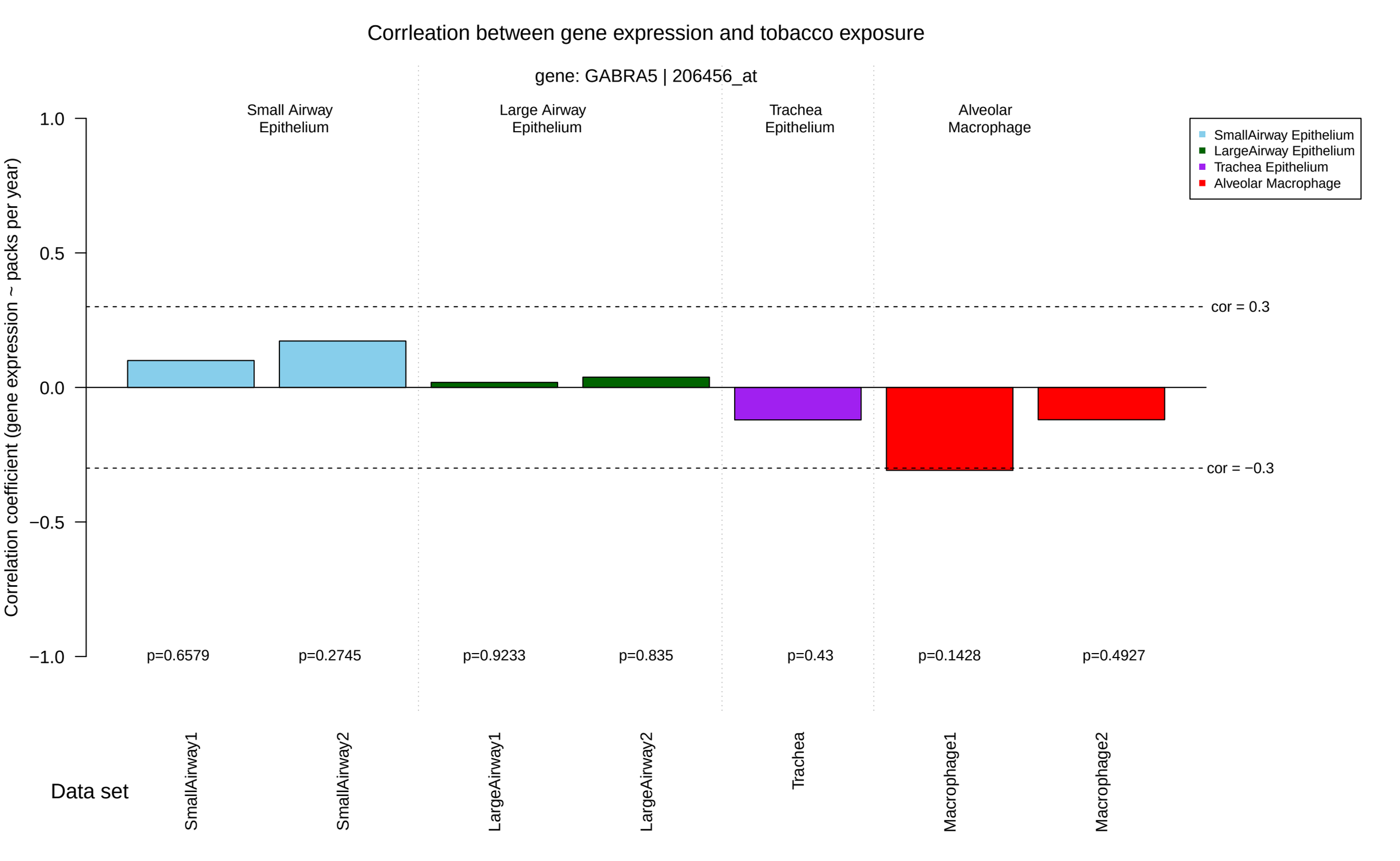

set: LargeAirway1 (p=0.4471)
GABRA5 | 206456_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8864)
GABRA5 | 206456_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2739)
GABRA5 | 206456_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2447)
GABRA5 | 206456_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2201)
GABRA5 | 206456_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRA5 | 206456_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6579
p=0.2745
p=0.9233
p=0.835
p=0.43
p=0.1428
p=0.4927
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2